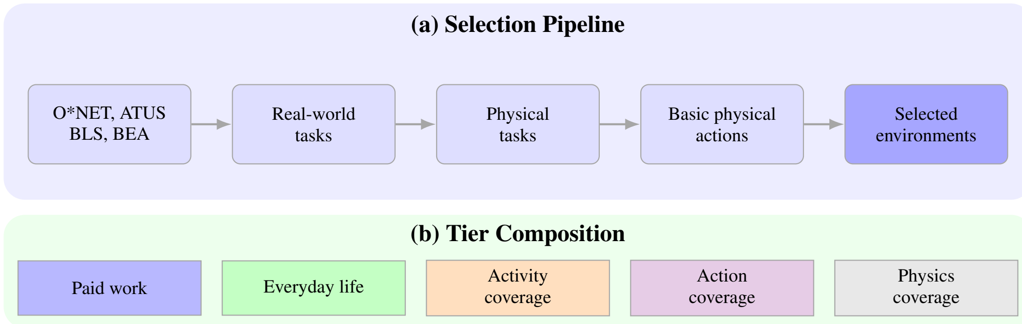

Figure 1: **Selection pipeline for robotics environments.** Starting from public data on paid work and everyday life, we identify physical tasks, break them into basic actions, and select environments that cover important and varied real-world activity.

2 Methodology

In this section, we introduce the problem setup and describe how we select robotics environments from real-world human activity.

Execution status. This draft reports the completed automated source, model, grouping, weighting, planning-feasibility, and tiered-selection runs. The planned human calibration, held-out evaluation, independent decomposition review, and human group review have not yet been run. Model-derived labels and planning assessments are therefore identified as such; they are not presented as human validation or empirical simulator calibration.

2.1 Problem Setup

Environment. We refer to an environment $\mathcal{E}$ as a simulated physical scene containing a robot, objects, and a specified initial state. The simulator advances the scene according to its physical model, while the agent acts through the robot's controls.

Task. A task $\mathcal{T} = (\mathcal{E}_{s_0}, p, V)$ consists of an environment $\mathcal{E}$ with initial state s_0, a natural-language instruction p, and a verification function V that maps the resulting trajectory to a score.

Interaction. At each step t, the agent receives an observation o_t from the simulated sensors and sends a robot action a_t. The simulator then advances to a new state and returns o_{t+1}. An episode begins from s_0 and its final score is determined by V.

2.2 Selecting Environments from Real-World Activities

Determining which physical tasks to include in a robotics benchmark is an important design decision. We use a simple principle: *prioritize physical activities that people actually perform.* We ground the selection in public U.S. data on paid work and everyday life (Figure 1). In a nutshell, we estimate the importance of occupations and everyday activities, identify tasks with a physical component, break those tasks into basic physical actions, group actions that require the same physical behavior, attribute activity importance to each action, filter to tasks that can be simulated faithfully, and apply tiered selection. We detail each step below.

Estimating the importance of activities. For paid work, we use O*NET occupations and tasks together with employment and wage data from the Bureau of Labor Statistics and national accounts from the Bureau of Economic Analysis. As in Gym-Anything, this gives each occupation an estimate of its share of economic activity. For everyday life, we use the American Time Use Survey to estimate how much population time is spent on activities such as cooking, cleaning, caregiving, exercise, recreation, and household maintenance. Where the Bureau of Economic Analysis provides a household-production estimate, we retain it as a separate measure. We do not add dollars and hours together; they remain separate views of importance.

Discovering physical tasks. For paid work, O*NET provides task statements for each occupation. For everyday life, the time-use categories are broader, so we use their official definitions and examples, together with web-grounded search when needed, to identify the concrete tasks people perform. We do not begin with a fixed list of robotics tasks. A task is kept when completing it depends on physical movement, contact with an object or person, use of a tool or machine, a change to a material, or observation from a physical position. The executed run cleans the task list by checking source records, removing duplicates, and applying an independent model-validation pass. Its uncertain decisions remain explicit because the planned human review has not yet been performed.

Breaking tasks into basic actions. Each physical task is broken into the smallest necessary actions that have a clear starting state and ending state. Alternative methods are kept separate: folding a shirt by hand and using a folding machine are not treated as one procedure. For each action, we record what acts, what is acted on, any tool that is used, the relevant contact or material behavior, and how completion can be checked. We call an action atomic when splitting it further would describe controller-level motion rather than another meaningful step in the task.

Grouping equivalent actions. The same action can appear in many tasks with different objects or tools. We group two actions when changing those details does not change the required state change, contact, or physical behavior. We keep them separate when the physics changes. For example, cutting a deformable solid and separating two fitted parts both divide one object state into two, but they require different contact and material models and therefore remain different actions.

Attributing importance to actions. Not every task is entirely physical, and a task with more described steps should not receive more weight simply because it was written in greater detail. We therefore allocate each activity's importance across its tasks and each task's physical portion across its actions:

$$\text{Weight}_{\text{action}} = \sum_{\text{activity}} \text{Weight}_{\text{activity}} \times s_{\text{task}} \times p_{\text{physical}} \times s_{\text{action}}. \tag{1}$$

Here, s_{task} is the task's share within an occupation or everyday-activity category, p_{physical} is the physical fraction of the task, and s_{action} is the action's share of the physical task. Task shares and action shares are normalized within their parent. We compute work and population-time weights separately. We retain household-production values as a third source measure, but do not allocate them to actions because the locked sources do not provide a complete activity-to-service crosswalk.

Filtering by simulation feasibility. Simulation feasibility is a planning screen rather than proof that every candidate has already been built. A candidate is selectable when there is a credible path to implement the required physics and assets, reset its starting state, measure its outcome, and test its fidelity against real data or a controlled experiment. Missing assets, custom physics, or calibration work place a candidate in `needs_development`; they do not reject it. A candidate remains nonselectable only when no credible implementation, reset, verification, or fidelity-test path is known.

Tiered selection. From the candidates that pass the planning screen, we select environments across tiers that balance importance with breadth: highest-weight paid work, highest-time everyday activities, coverage across occupation and time-use groups, important actions not yet covered, and physical behaviors not yet covered. Within each tier, we prefer environments that add the most remaining coverage while preserving the required fidelity target. The executed budget is 20/20/40/10/10 across the five tiers, for 100 newly selected environments (Appendix A).

A Environment Selection from Real-World Activities: Full Pipeline

This appendix provides the technical details and audited counts for the executed automated pipeline summarized in Section 2.2. The U.S. frame supplies the weights; international sources extend unweighted action discovery and grouping. All 100 selected grounding occurrences are U.S. records. Three planned human-review gates remain unexecuted, so the results below are an executed draft rather than a final human-validated study.

A.1 Phase 1: Activity Weight Calculation

Paid work. We assign an economic value to each O*NET occupation using the same scaling procedure as Gym-Anything:

1. **Wage bill**: compute employment $\times$ mean wage from BLS Occupational Employment and Wage Statistics.
2. **Labor compensation**: scale wage bills by the national ratio of total compensation to total wages from BEA accounts.
3. **Total economic activity**: scale labor compensation by the national ratio of GDP to compensation.

Everyday life. For each activity in the American Time Use Survey, we combine diary minutes with respondent survey weights to estimate population time. We retain the survey's activity code, location, and social context when available. Household-production values from BEA remain separate from population time. They are not allocated in this run because the locked BEA mapping covers only part of the activity frame.

The run begins from a source file that records the exact release, URL, retrieval date, and checksum for every input. Paid-work value, population time, and household-production value are stored in separate columns and are never summed.

Output: `activity_weights.csv` with the source, activity code, label, weight, unit, and source release for each occupation or everyday activity.

A.2 Phase 2: Physical Task Discovery

Paid-work tasks. For each occupation, we load its O*NET task statements and task ratings. Each occupation–task pair remains a separate record. Broader O*NET work-activity links are retained as context but do not create duplicate tasks.

Everyday-life tasks. For each time-use activity, we begin with the official label, definition, examples, and coding notes. When the category is too broad to describe a task, an LLM with web-search access proposes common concrete tasks. Each proposed task must be supported by an official source, manual, standard, or directly observable procedure. Categories are discovered freely rather than drawn from a fixed robotics taxonomy.

Physical-task identification. For each task, an LLM receives the source text and supporting evidence and returns whether the task has a physical component, the estimated physical fraction, the text or evidence that supports the decision, and the reason for excluding a nonphysical task. Mixed tasks are retained with a fractional estimate rather than forced into a yes/no label.

The executed discovery pass used `gemini-3.6-flash`, temperature zero, and low thinking. A separate validation pass used the same model at temperature zero with medium thinking. These are model decisions, not human physicality labels. Outputs are retained in the `raw_actions_v1` and `action_validation_v1` releases, with parallel international releases.

A.3 Phase 3: Task Catalog Cleanup

Three checks remove unsupported or duplicated records from the task catalog.

Source validation. Occupation codes, task identifiers, time-use codes, and source text are checked against the locked source releases. Join errors and exact duplicate rows are removed.

Task validation. For discovered everyday-life tasks, an independent pass checks that the task is a real and common interpretation of the source activity. Unsupported tasks are removed. For paid work, the pass checks that the physical interpretation follows from the O*NET task statement or its supporting evidence.

Physical-fraction review. The pre-execution plan specified a manually labeled development set, a sealed held-out evaluation, and review of high-importance or low-confidence decisions. The repository currently contains zero production human-review records, and the sealed holdout has not been opened. Human recall, precision, agreement, and manual-review sample counts are therefore not reported.

All model calls, search results used as evidence, and review decisions are saved for reproducibility.

A.4 Phase 4: Basic Action Discovery

For each retained physical task, an LLM receives the task statement, its evidence, the physical fraction, and any known objects, tools, and work context. It returns the ordered physical actions required to complete the task.

Each action contains: a short action description; the actor; the object or person acted on; any tool or support; the starting and ending state; contact before, during, and after the action; relevant material behavior; and a measurable completion condition. Alternative procedures are returned as separate branches. Reasoning, planning, waiting, and passive object motion are not recorded as agent actions.

The executed cleanup pass used `gemini-3.6-flash`, temperature zero, and medium thinking to check source support, repair action boundaries, and quarantine underspecified actions. The planned independent human decomposition review and held-out reconstruction evaluation have not yet been run. The release retains 33,281 clean occurrences; 16,161 are sufficiently specified for grouping and 17,120 remain in the source-underspecified quarantine.

A.5 Phase 5: Action Grouping

Action descriptions are first normalized for spelling and wording. Possible matches are then proposed using the action text, starting and ending states, object and tool roles, contact, and relevant physics.

Two actions are placed in the same group only if replacing their objects, tools, or scale preserves the intended state change, contact pattern, what must be controlled (such as position, force, or speed), material behavior, and completion condition. Similar verbs are not enough. Candidate pairs that differ in required physics remain separate.

Candidate pairs were generated with a minimum lexical-mechanics score of 0.24 and at most 8 neighbors per occurrence. Pair decisions used `gemini-3.6-flash`, temperature zero, and high thinking, followed by exact-signature compatibility and a negative-pair veto. No weights were used. The planned human review of ambiguous matches and high-weight groups has not yet been run. The verified catalog contains 92,215 candidate-pair decisions and 10,098 action groups.

A.6 Phase 6: Action Weight Attribution

For paid work, O*NET relevance, importance, and frequency ratings determine each task's share within its occupation. For everyday life, a broad time-use category's weight is divided among its supported concrete tasks. The physical fraction and action shares then allocate each task's weight according to Equation 1.

Task shares sum to one within an occupation or everyday-activity category, and action shares sum to one within each physical task. These constraints prevent repeated O*NET links, alternative procedures, or longer action lists from creating extra weight. We aggregate actions within the same group across all source tasks.

The output retains separate columns for paid-work economic value, population time, and household-production value. The pipeline reports uncertainty from task ratings, physical fractions, task shares, action shares, and disputed groupings rather than collapsing the result to one unsupported ranking.

Physical and action shares were estimated with `gemini-3.6-flash`, temperature zero, and high thinking. Unresolved shares remain missing rather than being imputed. Output: `action_weights.csv`, with each action group, its separate market-work and population-time weights, uncertainty, provisional rank, and contributing source tasks. The household-production column remains explicitly unallocated.

A.7 Phase 7: Simulation Feasibility Evaluation

For each action group, we evaluate:

- whether the simulator can represent the required contact, material behavior, and other physics at the needed fidelity;
- whether a suitable robot, sensor setup, objects, tools, and scene can be built;
- whether the starting state can be reset and the completion condition can be measured;
- whether the environment can run within the available compute budget; and
- what real-world data or experiment is needed to set and test the physical parameters.

Each candidate is classified as `ready`, `needs_development`, or `not_currently_supported`, with a written reason. Both `ready` and `needs_development` are selectable under the permissive credible-path rule. These are planning labels derived from the catalog signature and a representative occurrence, not evidence that the candidate is implemented or calibrated.

Output: `simulation_feasibility.jsonl.gz`, containing the required physics, assets, measurements, expected compute, decision, and reason.

A.8 Phase 8: Tiered Selection

An environment is selectable if there is a credible path to implement its required behavior, reproduce its initial state, verify its outcome, and test its fidelity against real data or a controlled experiment. Missing implementation or calibration work alone does not make it nonselectable.

Selection proceeds across five tiers:

Tier	Budget	Strategy
k1 (Paid work)	20	Highest paid-work weight overall
k2 (Everyday life)	20	Highest population time; household-production value unavailable in this run
k3 (Activity coverage)	40	Round-robin across occupation groups and time-use categories
k4 (Action coverage)	10	Important basic actions not covered by earlier tiers
k5 (Physics coverage)	10	Required physical behaviors not covered by earlier tiers

Table 1: Executed tier budgets for the 100-environment planned portfolio.

Within a tier, candidates are ranked by the remaining activity weight and basic actions they add. A candidate receives planning credit only when its catalog action has a credible implementation, reset, verifier, and fidelity-test path. This is not implementation or calibration credit. Unselected candidates remain listed as uncovered rather than being replaced by easier tasks.

Output: `selected_environments.json`, containing each selected environment, its tier, covered actions and activities, required physics, feasibility evidence, and selection reason.

A.9 Pipeline Statistics

Metric	Value
Weighted U.S. source units	19,261
O*NET occupations	1,016
ATUS activity categories	465
Physical source units (U.S.; all frames)	6,808; 18,560
Clean action occurrences (U.S.; all frames)	12,238; 33,281
Sufficient occurrences grouped	16,161
Action groups after matching	10,098
Planning candidates (all selectable)	10,098
Planning classes (ready; needs development)	3,303; 6,795
New environments selected	100
Activity buckets covered	37
Physical regimes covered	20

Table 2: Audited automated-pipeline statistics. "All frames" includes unweighted international discovery sources; all selected grounding occurrences are U.S. records. Feasibility counts are planning assessments.